

SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

LAB EXPERIMENTS

ECA1011 Microprocessors and Microcontrollers

EXPERIMENT 1

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INDEX

S.No	Date	Title of Experiment	Page No
1.	15/10/26	EX 1: 8-bit and 16 bit Arithmetic Operations	
2.	15/10/26	EX 2: Logical, Bit-manipulation And Shift Operations	
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			

DATE: 15/10/2026

EXPERIMENT 01: ANALYSIS, DEVELOPMENT AND VERIFICATION OF 8-BIT AND 16-BIT ARITHMETIC OPERATIONS USING ASSEMBLY LANGUAGE PROGRAM

Objective:

1. To analyze, develop & verify 8-bit Addition and Subtraction
2. To analyze, develop & verify 16-bit Addition and Subtraction
3. To analyze, develop & verify 8-bit Multiplication and Division
4. To analyze, develop & verify 16-bit Multiplication and Division

Components Required:

8085 Simulator or 8085 Intel Microprocessor

Procedure:

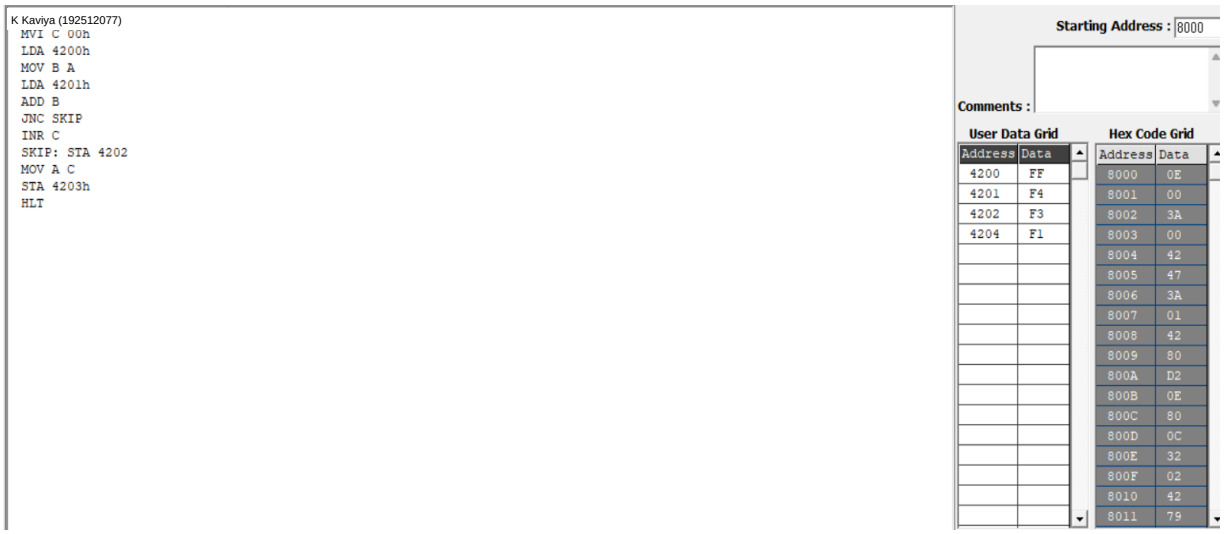
i. Switch ON the 8085-microprocessor kit or open the 8085 simulator.
ii. Enter the program opcodes into the specified starting memory address using the memory examine/substitute command.
iii. Store the required input data in the specified memory locations, if provided.
iv. Execute the program from the given starting address using the GO/EXEC command.
v. Examine the memory locations assigned for the results.
vi. Record the output values and verify them with the expected arithmetic results.
vii. Repeat the procedure for all four programs and record the observations.

1. Analysis of 8-bit Addition and Subtraction :

PROGRAM [Addition]

```
MVI C 00h
LDA 4200h
MOV B A
LDA 4201h
ADD B
JNC SKIP
INR C
SKIP : STA 4202h
MOV A C
STA 4203h
HLT
```

OUTPUT

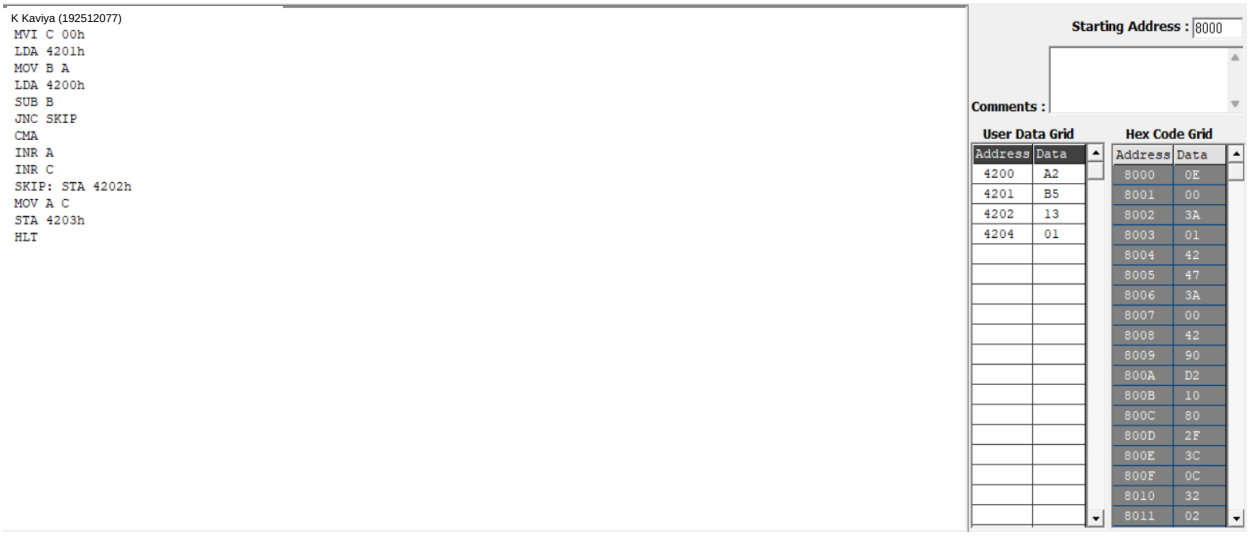


Memory	Data
4200	FF
4201	F4
4202	F3
4203	01

PROGRAM [Subtraction]

```
MVI C 00h
LDA 4201h
MOV B A
LDA 4200h
SUB B
JNC SKIP
CMA
INR A
INR C
SKIP : STA 4202h
MOV A C
STA 4203h
HLT
```

OUTPUT



Memory	Data
4200	A2
4201	B5
4202	13
4203	01

2. Analysis of 16-bit Addition and Subtraction:

PROGRAM [Addition]

```
LHLD 4510
XCHG
LHLD 4512
DAD D
SHLD 4514
MVI A 00
JNC SKIP
MVI A 01
SKIP: STA 4516
HLT
```

OUTPUT

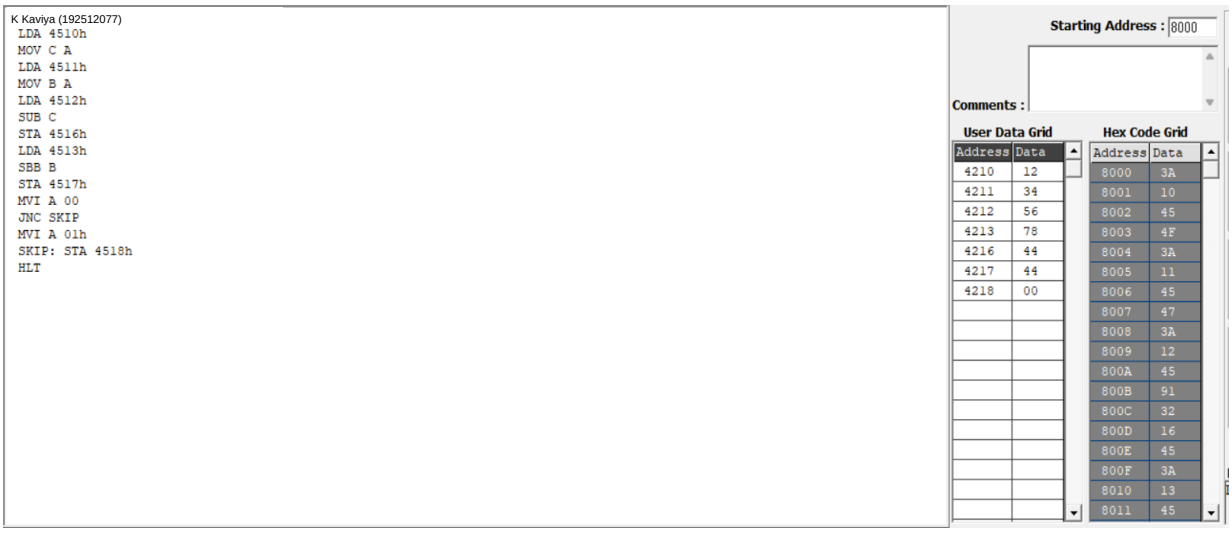


Memory	Data
4210	12
4211	34
4212	56
4213	78
4216	68
4217	AC
4218 (carry)	00

PROGRAM [Subtraction]

```
LDA 4510
MOV C  A
LDA 4511
MOV B  A
LDA 4512
SUB C
STA 4516
LDA 4513
SBB B
STA 4517
MVI A 00H
JNC SKIP
MVI A  01
SKIP: STA 4518
HLT
```

OUTPUT



Memory	Data
4210	12
4211	34
4212	56
4213	78
4216	44
4217	44
4218 (carry)	00

3. Analysis of 8-bit Multiplication and Division:

PROGRAM [Multiplication]

```
LXI H 0000H
LDA 4201h
MOV E A
LDA 4202
MOV C A
REPEAT : DAD D
DCR C
JNZ REPEAT
SHLD  4203
HLT
```

OUTPUT



Memory	Data
4201	1A
4202	C4
4203	E8
4204	13

PROGRAM [Division]

```
LDA 4201h
MOV E A
LDA 4200h
MOV B A
MOV A E
MVI C 00h
AGAIN: CMP B
JC STORE
SUB B
INR C
JMP AGAIN
STORE: STA 4203h
MOV A C
STA 4202h
HLT
```

OUTPUT



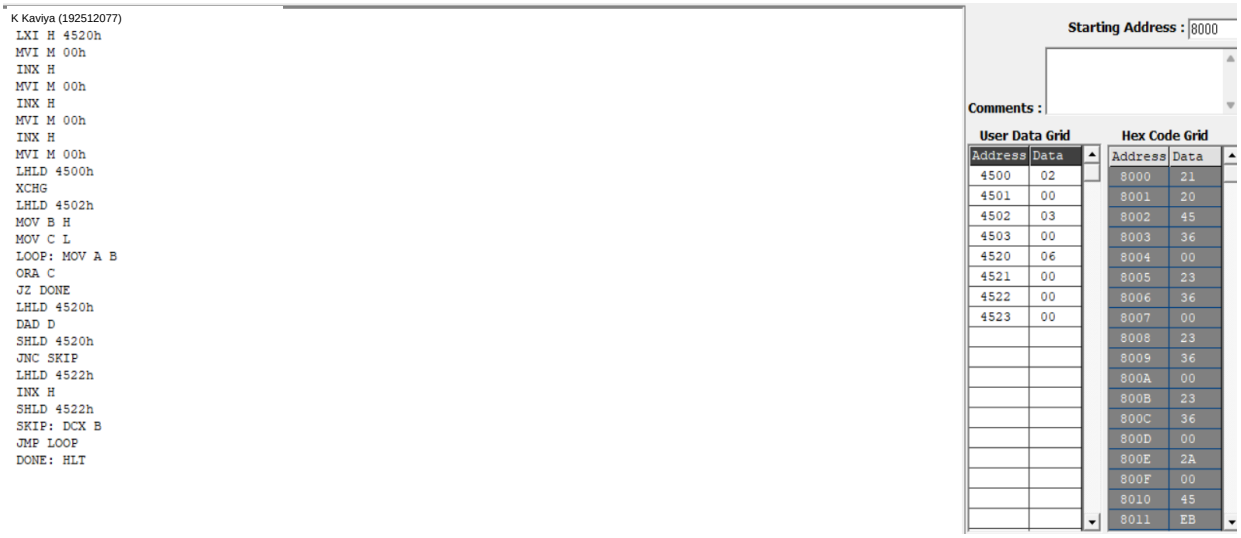
Memory	Data
4200	4A
4201	5B
4202	01
4203	10

4. Analysis of 16-bit Multiplication and Division:

PROGRAM [Multiplication]

```
LXI H 4520
MVI M 00
INX H
MVI M 00
INX H
MVI M 00
INX H
MVI M 00
LHLD 4500
XCHG
LHLD 4502
MOV B H
MOV C L
LOOP: MOV A B
ORA C
JZ DONE
LHLD 4520
DAD D
SHLD 4520
JNC SKIP
LHLD 4522
INX H
SHLD 4522
SKIP: DCX B
JMP LOOP
DONE: HLT
```

OUTPUT

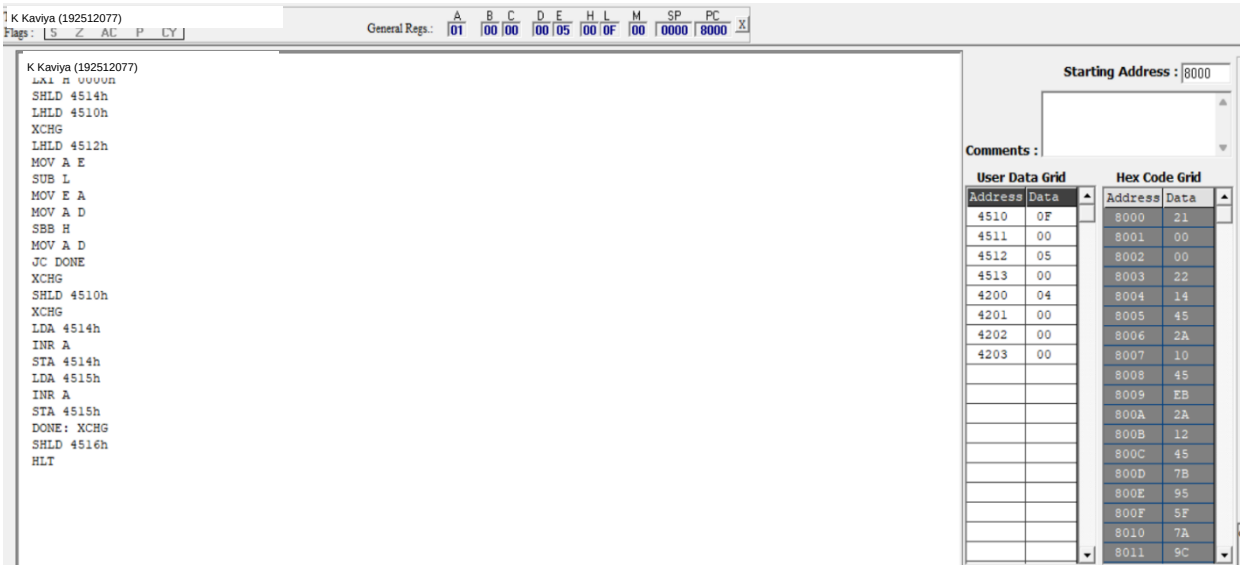


Memory	Data
4500	02
4501	00
4502	03
4503	00
4520	06
4521	00
4522	00
4523	00

PROGRAM [Division]

```
LXI H 0000
SHLD 4514
LHLD 4510
XCHG
DIVLOOP:
LHLD 4512
MOV A E
SUB L
MOV E A
MOV A D
SBB H
MOV D A
JC DONE
XCHG
SHLD 4510
XCHG
LDA 4514
INR A
STA 4514
JNZ DIVLOOP
LDA 4515
INR A
STA 4515
JMP DIVLOOP
DONE:
XCHG
SHLD 4516
HLT
```

OUTPUT



Memory	Data
4510	14
4511	00
4512	05
4513	00
4200	04
4201	00
4202	00
4203	00

Result:

Thus, 8-bit and 16-bit addition, subtraction, multiplication, and division programs were executed successfully using the 8085 microprocessors, and the results were verified by examining the specified memory locations and flag conditions.

DATE: 15/10/2026

EXPERIMENT 02: LOGICAL, BIT-MANIPULATION AND SHIFT OPERATIONS USING 8-BIT AND 16-BIT DATA

Objective:

1. To analyze, develop and verify logical AND, OR and XOR operations.
2. To analyze, develop and verify one's and two's complement operations on 8-bit and 16-bit data.
3. To analyze, develop and verify bit setting, clearing, toggling, rotate and shift operations.
4. To verify the output results and carry flag using the 8085 simulator.

Components Required:

8085 Simulator

Procedure:

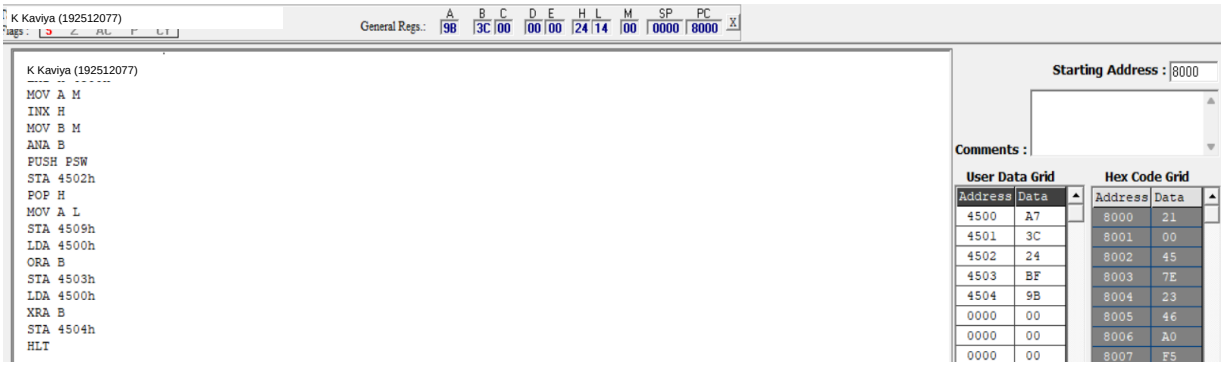
i) Switch ON the 8085-microprocessor trainer kit.
ii) Enter the program opcodes at the specified starting memory address using the memory examine/substitute command.
iii) Store the required input data in the specified memory locations.
iv) Execute Program A from 4100H and Program B from 4200H using the GO/EXEC command.
v) Examine the memory locations assigned for the output results and flag byte.
vi) Record the output values and verify them against the expected hexadecimal results.
vii) Repeat the procedure for both programs and record the observations.

1. Analysis of Logical Operations

PROGRAM [AND,OR,XOR in 8-bit]

```
LXI H, 4500H
MOV A, M
INX H
MOV B, M
ANA B
PUSH PSW
STA 4502H
POP H
MOV A, L
STA 4509H
LDA 4500H
ORA B
STA  4503H
LDA  4500H
XRA B
STA  4504H
HLT
```

OUTPUT



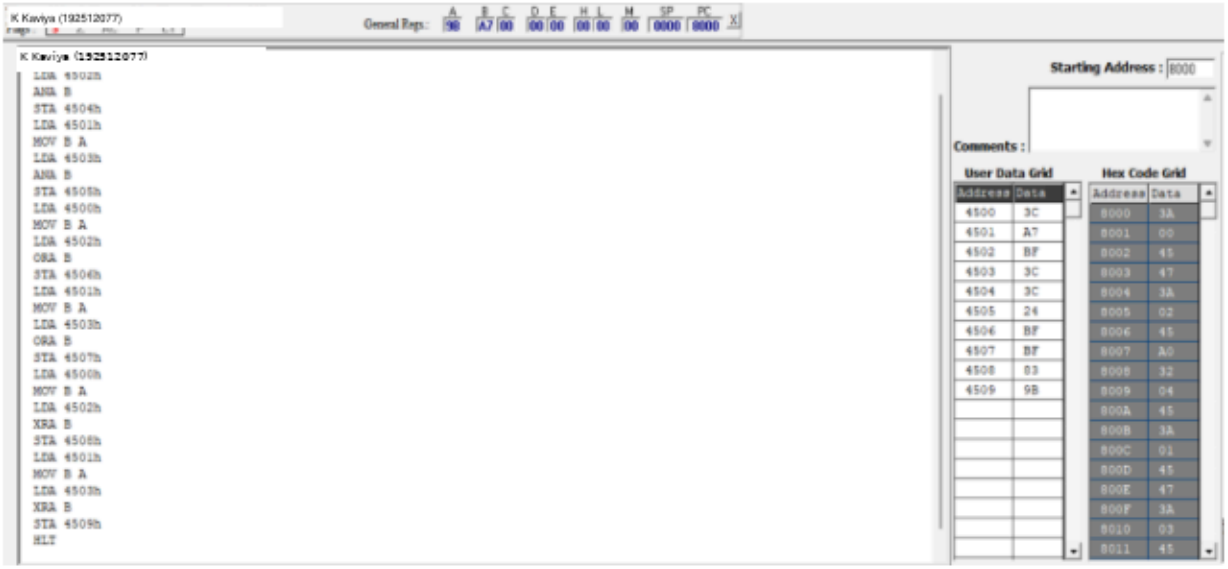
Memory	Data
4500H	A7H

4501H	3CH
4502H	24H
4503H	BFH
4504H	9BH

PROGRAM [AND,OR,XOR in 16-bit]

```
LDA 4500
MOV B A
LDA 4502
ANA B
STA 4504
LDA 4501
MOV B A
LDA 4503
ANA B
STA 4505
LDA 4500
MOV B A
LDA 4502H
ORA B
STA 4506
LDA 4501
MOV B A
LDA 4503
ORA B
STA 4507
LDA 4500
MOV B A
LDA 4502
XRA B
STA 4508
LDA 4501
MOV B A
LDA 4503
XRA B
STA 4509
HLT
```

OUTPUT



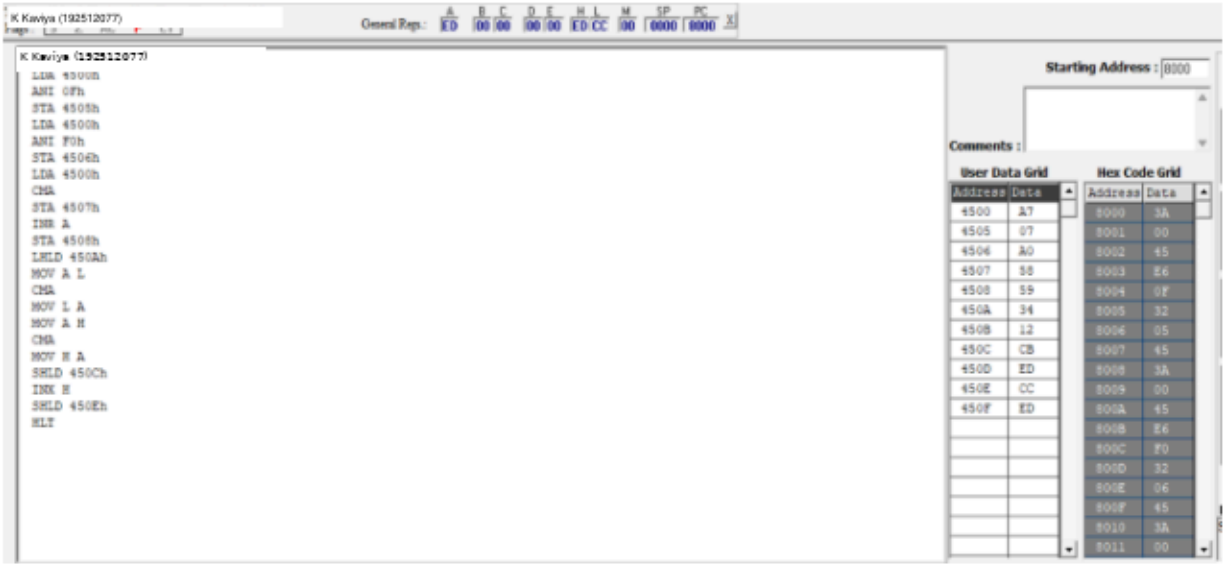
Memory	Data
4500	3CH
4501	A7H
4502	BFH
4503	3CH
4504	3CH
4505	24H
4506	FFH
4507	BFH
4508	83H
4509	9BH

2. Analysis of One's and Two's Complement

PROGRAM [16-bit]

```
LDA 4500H
ANI 0FH
STA 4505H
LDA 4500H
ANI F0H
STA 4506H
LDA 4500H
CMA
STA 4507H
INR A
STA 4508H
LHLD 450AH
MOV A, L
CMA
MOV L, A
MOV A, H
CMA
MOV H, A
SHLD 450CH
INX H
SHLD 450EH
HLT
```

OUTPUT



Memory	Data
4500	A7
4505	07
4506	A0
4507	58
4508	59
450A	34
450B	12
450C	CB
450D	ED
450E	CC
450F	ED

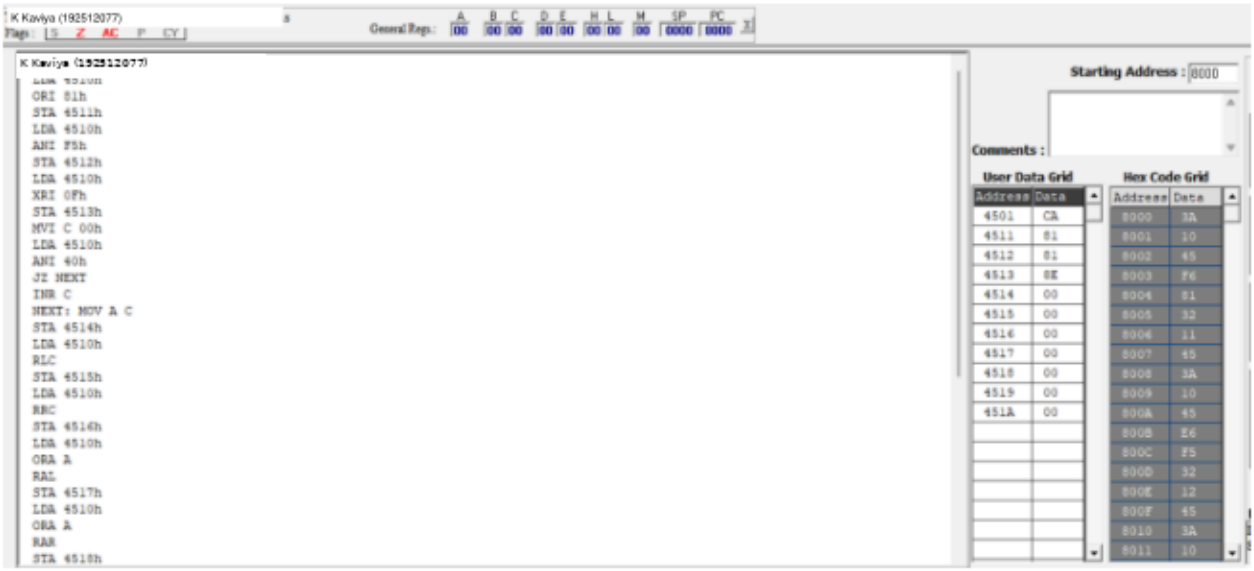
3. Analysis of Bit Setting, Clearing, Rotate

PROGRAM [Bit Manipulation]

```
LDA 4510H
ORI 81H
STA 4511H
LDA 4510H
ANI F5H
STA 4512H
LDA 4510H
XRI 0FH
STA 4513H
MVI C, 00H
LDA 4510H
ANI 40H
JZ NEXT
INR C
NEXT: MOV A, C
STA  4514H
LDA  4510H
RLC
STA  4515H
LDA  4510H
RRC
STA  4516H
LDA  4510H
ORA A
RAL
STA  4517H
LDA  4510H
ORA A
RAR
STA 4518H
LDA 4510H
MOV B, A
RAL
MOV A, B
RAR
STA  4519H
LDA  4510H
MVI  B,  08H
MVI  C,  00H
CNT: RLC
```

```
JNC SKIP
INR C
SKIP: DCR B
JNZ CNT
MOV A, C
STA 451AH
HLT
```

OUTPUT



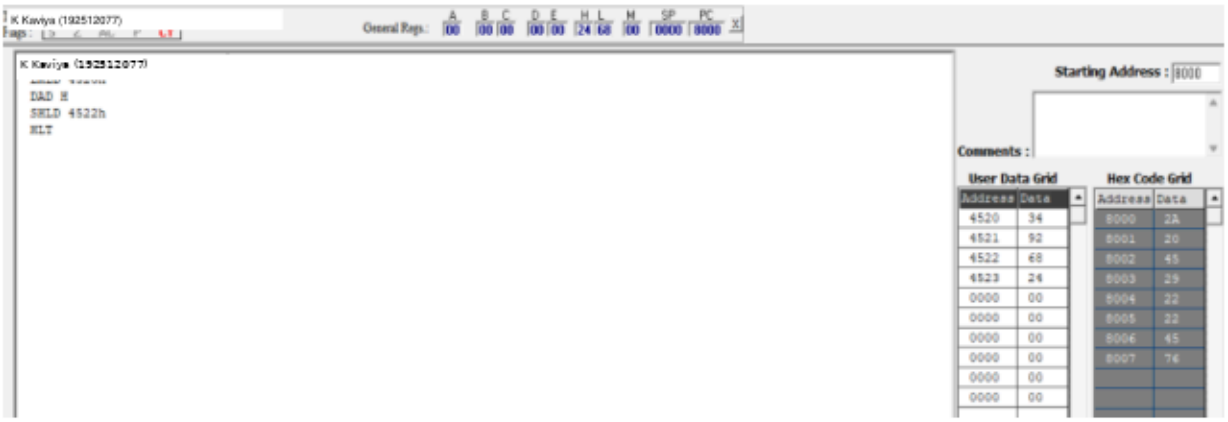
Memory	Data
4510H	CAH
4511H	CBH
4512H	C0H
4513H	C5H
4514H	01H
4515H	95H
4516H	65H
4517H	94H
4518H	65H
4519H	E5H
451AH	04H

4. Analysis of Shift Operations

PROGRAM [16-bit Shift Left]

```
LHLD 4520H
DAD H
SHLD 4522H
HLT
```

OUTPUT



Memory	Data
4520H	34H
4521H	92H
4522H	68H
4523H	24H

Result:

Thus, the logical, bit-manipulation, complement, rotate and shift operations were performed using the 8085 simulator. The output values were examined in memory and verified against the expected hexadecimal results.